QP CODE: **Reg No:.....**

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2026

Fifth Semester

MG5DSECFS306 - MOBILE AND WIRELESS SECURITY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions.

Each question carries **2** marks

1. Identify two commonly used wireless hardware devices**. [U] [CO1]**
2. Relate wireless network protocols to their roles in wireless communication. **[U] [CO1]**
3. Identify two important limitations of WEP security. **[U] [CO1]**
4. Explain the importance of encryption in protecting information in wireless communication**. [U] [CO2]**
5. Describe the importance of non-repudiation in network security. **[U] [CO2]**
6. Select an appropriate security mechanism for protecting confidential wireless communication. **[A] [CO2]**
7. Explain the purpose of a Virtual Private Network (VPN) in network security. **[U] [CO2]**
8. Examine the basic features of IEEE 802.11b. **[An][CO3]**
9. Analyse the main parameters of WLAN configuration. **[An][CO3]**
10. Examine the purpose of a WEP decryption script. **[An][CO3]**
11. Examine the role of Global Mobile Satellite Systems in mobile communication. **[An][CO4]**
12. Compare wireless enterprise networks with wired networks based on two advantages. **[An][CO4]**

(2 × 10 = 20)

Part B

Short Answer Questions

Answer any **6** questions.

Each question carries **5** marks

13. Describe the characteristics of different wireless network protocols. **[U] [CO1]**
14. Explain the differences between wireless threats and wireless attacks with suitable examples. **[U][CO1]**
15. Explain suitable physical hardware security measures for protecting a wireless network. **[U][CO2]**
16. Organize identification, authentication, and authorization according to their roles in network security. **[A][CO2]**
17. Implement suitable mitigation and protection measures for attacks and vulnerabilities as part of the security analysis process. **[A][CO2]**
18. Differentiate between the physical layer and media access frame format of IEEE 802.11. **[An][CO3]**
19. Compare different types of WEP attacks. **[An][CO3]**
20. Categorize the main components of Bluetooth protocols according to their functions. **[An][CO4]**

(5 × 6 = 30)

Part C

Essay Questions

Answer any **2** questions.

Each question carries **10** marks

21. Describe airborne viruses and explain their effects on wireless devices and networks. **[U][CO1]**
22. Organize the different security solutions into standards and policy solutions, network solutions, and software solutions. **[A][CO2]**
23. Examine the structure and operation of wireless enterprise networks. **[An][CO4]**

(10 × 2 = 20)